

Good job

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IIT Madras BS DEGREE

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Roll No. 21F1001107

Subject Linear Statistical Models

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Exam Session: Forenoon / Afternoon

No. of sheets used 8

No. of pages used 15

Paper+Pen Exams – Instructions

(Please read these instructions carefully before starting the exam.)

1. Questions will be provided on the computer. Blue coloured page is the cover sheet where you have to fill your details. DO NOT write your answers in this Blue coloured sheet.
2. Yellow coloured pages with IIT Madras B.S. Degree header will be given to each candidate as Answer sheets. You have to mark/write on these Yellow -coloured answer sheets, tie the sheets together and return the same to the invigilator.
3. Answers written on any other paper will not be evaluated.
4. You have to write your Name & Roll Number in all Answer Sheets. Answer Sheets without your name and roll number will not be evaluated.
5. Tying the Answer Sheets securely is the candidate's responsibility.
6. Please DO NOT bring any type of Calculator for use in the exam. On-screen calculator will be available during the exam.
7. Rough paper will be provided to candidate for rough work. Candidates have to write their Name and Roll number on the rough paper before they start using it. These rough papers must be returned to the invigilator after the end of the examination.
8. Course Specific instructions, wherever applicable, will be displayed on your monitor.
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- The End -



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LSM

(Q1)

✓ 2

(a) ~~option~~ option are correct one only.

why?

(a) $\bar{X} \sim N(\mu, (\sigma/\sqrt{n})^2)$

therefore $\frac{(X-\mu)}{(S/\sqrt{n})} \sim t_{n-1}$ [T-test]

(b) $\bar{X} \sim N(\mu, \frac{\sigma^2}{n})$ not $(\mu, n\sigma^2)$

(c) & (d)

$\sum_{i=1}^4 (3X_i + 5)$

$$\begin{aligned} E\left(\sum_{i=1}^4 3X_i + 5\right) &= 3E\left(\sum_{i=1}^4 X_i\right) + 4 \cdot 5 = 3E\left(\sum_{i=1}^4 X_i\right) + 20 \\ &= 3 \sum_{i=1}^4 E(X_i) + 20 \\ &= 3 \cdot 4 \cdot \mu + 20 = 12\mu + 20 \end{aligned}$$

$$\begin{aligned} (d) \text{Var}\left(\sum_{i=1}^4 (3X_i + 5)\right) &= \text{Var}\left(\sum_{i=1}^4 3X_i\right) \\ &= 9 \text{Var}\left(\sum_{i=1}^4 X_i\right) = 9 \cdot \sum_{i=1}^4 \text{Var}(X_i) \\ &= 9 \cdot 4 \cdot \sigma^2 = 36\sigma^2 \\ \text{② } \sum_{i=1}^4 (3X_i + 5) &\sim N(12\mu + 20, 36\sigma^2) \end{aligned}$$

Only option (a) is correct

(Q2)

(a) "ANOVA"

is the correct answer 2
to ~~too~~ compare the difference in
sample means of 3 or more group
we use ANOVA

Z & t test are used for 2 group
sample mean difference.

Z $\rightarrow$ when pop. var. is given
t $\rightarrow$ " " " not given
& we use sample variance.

χ^2 - test $\rightarrow$ Used for Goodness of fit
or Test of Independence.

ANOVA:

$$H_0: \mu_1 = \mu_2 = \mu_3 \dots = \mu_n$$

$\Rightarrow$ Sample mean for all groups are
same

H_a : at least one pair of sample
means is different.



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(Q3)

No, the analyst's choice is 2 not correct in my ~~of~~ opinion as we use one-way anova when only one parameter is changing and we want to observe if the difference in sample means is because of that particular parameter only.

Since here in the scenario we have 2-factors:

→ Fertilizer type (3)

→ Irrigation type (2)

Therefore we will have different combinations ($6 = 3 \times 2$). And therefore we will have to use 2-way Anova as it will take both Fertilizer & Irrigation into consideration.

(Qu)(i)

$$\bar{Y}_{10} = \frac{2+3+3+4}{4} = 3$$

$$\bar{Y}_{20} = \frac{5+4+6+5}{4} = 5$$

$$\bar{Y}_{30} = \frac{3+2+4+3}{4} = 3$$

$$\bar{Y}_{40} = \frac{6+5+5+4}{4} = 5$$

$$\bar{Y}_{10}, \bar{Y}_{20}, \bar{Y}_{30}, \bar{Y}_{40} = 3, 5, 3, 5$$

✓ (2)

(ii)

$$\bar{Y}_{00} = \frac{2+3+3+4+5+4+6+5+3+2+4+3+6+5+5+4}{16} = \frac{24+40}{16} = 4$$

~~$$SSB = \sum_{i=1}^K n_i (\bar{Y}_{i0} - \bar{Y}_{00})^2$$~~

$$SSB = \sum_{i=1}^K n_i (\bar{Y}_{i0} - \bar{Y}_{00})^2$$

$$= 4(3-4)^2 + 4(3-5)^2 + 4(3-4)^2 + 4(3-5)^2$$

$$= 4+4+4+4 = 16$$

$$SSB = 16$$



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$$SSW = \sum_{i=1}^k \sum_{j=1}^{n_i} (Y_{ij} - \bar{Y}_{i.})^2$$

$$= (2-3)^2 + (3-3)^2 + (3-3)^2 + (4-3)^2 \\ + (5-5)^2 + (4-5)^2 + (6-5)^2 + (5-5)^2 \\ + (3-3)^2 + (2-3)^2 + (4-3)^2 + (3-3)^2 \\ + (6-5)^2 + (5-5)^2 + (5-5)^2 + (4-5)^2$$

$$= (1^2 + 1^2) \times 4 = 2 \times 4 \\ = 8$$

$$SSW = 8$$

4

(ii) $SSW = 8$ & $SSB = 16$

(iii) $MSB = \frac{SSB}{K-1} = \frac{16}{4-1} = \frac{16}{3} = 5.333$

2

$$MSW = \frac{SSW}{n-K} = \frac{8}{16-4} = \frac{8}{12} = 0.666$$

(i) w)

 H_0 :-

$$\mu_{D_1} = \mu_{D_2} = \mu_{D_3} = \mu_{D_4}$$

$\Rightarrow$ Average wt. loss due to all diet types is same. (2)

 H_a :-

Atleast Any ~~1~~ pair of average wt. loss due to diet is different.

(v) For Hypothesis testing we do "F test".

Under H_0 :

$$\frac{MSB}{MSW} \sim F_{(k-1, n-k)}$$

$$= \left(\frac{16}{3}\right) \div \left(\frac{8}{12}\right) \sim F(3, 12)$$

$$\Rightarrow \frac{16^2}{3} \times \frac{12^4}{8} = .8 \sim F(3, 12)$$

$$F_{\text{calculated}} = .8$$

$$F_{\text{Tabulated}} = 3.49 \quad (\text{at } \alpha = 0.05)$$

$$F_{(3, 12)} = 3.49$$



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Since

$$F_{\text{tabulated}} < F_{\text{calculated}}$$



We reject H_0 (null hypothesis)
& conclude that : 4

Average weekly wt. loss (kg)
is not same across all four
Diet programs and atleast one
pair has different avg. weekly
wt. loss.

(vi)	Sum Sq.	Dof	M.S.	F _{stat}	3
Treatment	16	3	5.33	8	
Error	8	12	0.666		
Total	24	15			

⑤

$$Y = \beta x_i + \epsilon_i \quad i = 1, 2, \dots, n$$

$$\epsilon_i \sim N(0, \sigma^2)$$

(i) Given

$$\hat{\beta}_1 = \frac{\sum x_i y_i}{\sum x_i^2}, \quad \hat{\beta}_2 = \frac{\sum y_i}{\sum x_i}$$

$$\hat{\beta}(a, b) = a \hat{\beta}_1 + b \hat{\beta}_2$$

for $\hat{\beta}(a, b)$ to be unbiased:

$$E(\hat{\beta}(a, b)) = \beta$$

Now,

$$E(\hat{\beta}(a, b)) = a E(\hat{\beta}_1) + b E(\hat{\beta}_2) \quad \left| \begin{array}{l} \text{linearity of} \\ E \end{array} \right.$$

Now,

$$E(\hat{\beta}_1) = E\left(\frac{\sum x_i y_i}{\sum x_i^2}\right)$$

$$= \frac{1}{\sum x_i^2} E(\sum x_i y_i)$$

x_i are constant
 $\Rightarrow \sum x_i^2 = \text{const}$

$$= \frac{1}{\sum x_i^2} E(\sum x_i (\beta x_i + \epsilon_i))$$

$$= \frac{1}{\sum x_i^2} E(\sum \beta x_i^2 + \sum x_i \epsilon_i)$$



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$$= \frac{1}{\sum x_i^2} \left(E(\beta \sum x_i^2) + E(\sum x_i \epsilon_i) \right)$$

$$= \frac{1}{\sum x_i^2} \left(\beta E(\sum x_i^2) + \sum E(x_i \epsilon_i) \right)$$

$$= \frac{1}{\sum x_i^2} \left(\beta \sum x_i^2 + \sum x_i E(\epsilon_i) \right) \quad \left| \begin{array}{l} \sum x_i^2 \text{ is constant} \\ E(c) = c \end{array} \right.$$

Since $\epsilon_i \sim N(0, \sigma^2)$

$$E(\epsilon_i) = 0$$

$$\Rightarrow E(\hat{\beta}_1) = \frac{\beta \sum x_i^2}{\sum x_i^2} = \beta$$

Now,

$$E(\hat{\beta}_2) = E\left(\frac{\bar{y}}{\bar{x}}\right) = E\left(\frac{\sum y_i / n}{\sum x_i / n}\right) = E\left(\frac{\sum y_i}{\sum x_i}\right)$$

Since, $x_i = \text{constant} \Rightarrow \sum x_i = \text{constant}$

$$\begin{aligned} \Rightarrow E(\hat{\beta}_2) &= \frac{1}{\sum x_i} E(\sum y_i) = \frac{1}{\sum x_i} E(\sum (x_i \beta + \epsilon_i)) \\ &= \frac{1}{\sum x_i} E(\sum (x_i \beta + \epsilon_i)) \end{aligned}$$

$$\begin{aligned}
 &= \frac{1}{\sum x_i} \left(E(\sum x_i \beta) + E(\sum \epsilon_i) \right) \\
 &= \frac{1}{\sum x_i} \left(\beta \sum E(x_i) + \sum E(\epsilon_i) \right) \\
 &= \frac{1}{\sum x_i} \left(\beta \sum x_i + 0 \right) \\
 &= \beta \frac{\sum x_i}{\sum x_i} = \beta
 \end{aligned}$$

So,

$$E(\hat{\beta}_1) = E(\hat{\beta}_2) = \beta$$

Therefore

$$\begin{aligned}
 E(\hat{\beta}(a,b)) &= a E(\hat{\beta}_1) + b E(\hat{\beta}_2) \\
 &= a\beta + b\beta \\
 &= (a+b)\beta
 \end{aligned}$$

but we want $\hat{\beta}(a,b)$ to be unbiased

$$\Rightarrow E(\hat{\beta}(a,b)) = \beta$$

$$\Rightarrow (a+b) = 1 \quad (\text{by comparison})$$

Therefore

$$a + b = 1$$

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(ii)

$$\text{Var}(\hat{\beta}_1) = \text{Var}\left(\frac{\sum x_i y_i}{\sum x_i^2}\right)$$

$$= \frac{1}{(\sum x_i^2)^2} \text{Var}(\sum x_i y_i)$$

$$\left| \sum x_i^2 = \text{constant} \right.$$

$$= \frac{1}{(\sum x_i^2)^2} \text{Var}(\sum x_i (\beta x_i + \epsilon_i))$$

$$= \frac{1}{(\sum x_i^2)^2} \text{Var}(\sum \beta x_i^2 + \sum x_i \epsilon_i)$$

$$= \frac{1}{(\sum x_i^2)^2} \text{Var}(\sum x_i \epsilon_i)$$

$$\left| \begin{array}{l} \sum \beta x_i = \text{constant} \\ \text{Var}(x + \text{const}) \\ = \text{Var}(x) \end{array} \right.$$

$$= \frac{1}{(\sum x_i^2)^2} \sum \text{Var}(x_i \epsilon_i)$$

$$\left| \begin{array}{l} \sum \text{Var}(x_i) \\ = \text{Var}(\sum x_i) \\ \text{for iid samples} \end{array} \right.$$

$$= \frac{1}{(\sum x_i^2)^2} \sum x_i^2 \text{Var}(\epsilon_i)$$

$$= \frac{\sigma^2 \sum x_i^2}{(\sum x_i^2)^2} = \frac{\sigma^2}{\sum x_i^2}$$



$$\text{Var}(\hat{\beta}_1)$$

$$= \frac{\sigma^2}{\sum x_i^2}$$

$\hat{\beta}_1$ is Least Sq estimator (without the intercept)

$$\begin{aligned} \text{Var}(\hat{\beta}_2) &= \text{Var}\left(\frac{\bar{y}}{\bar{x}}\right) = \text{Var}\left(\frac{\sum y_i}{\sum x_i}\right) \\ &= \frac{1}{(\sum x_i)^2} \text{Var}(\sum y_i) = \frac{1}{(\sum x_i)^2} \text{Var}(\sum x_i \beta + \sum \epsilon_i) \\ &= \frac{1}{(\sum x_i)^2} \text{Var}(\sum \epsilon_i) \quad | \quad \sum x_i \beta = \text{constant} \\ &= \frac{1}{(\sum x_i)^2} \sum \text{Var}(\epsilon_i) = \frac{1}{(\sum x_i)^2} \sum \sigma^2 = \frac{n\sigma^2}{(\sum x_i)^2} \\ &= \frac{\sigma^2}{n(\bar{x})^2} \end{aligned}$$

Pls. look at extra work in the end as well

$$\text{Var}(\hat{\beta}_1) < \text{Var}(\hat{\beta}_2)$$

as $\hat{\beta}_1$ is the BLUE estimator according to Gauss-Markov
OLS estimator is BLUE
& $\hat{\beta}_1$ is OLS estimator.

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$$\text{Var}(\hat{\beta}(a, b))$$

$$= \text{Var}(a\hat{\beta}_1 + b\hat{\beta}_2)$$

$$= a^2 \text{Var}(\hat{\beta}_1) + b^2 \text{Var}(\hat{\beta}_2) + 2ab \text{Cov}(\hat{\beta}_1, \hat{\beta}_2)$$

$$= a^2 \frac{\sigma^2}{\sum x_i^2} + \frac{b^2 n \sigma^2}{(\sum x_i)^2} + 2ab \left(\frac{\sigma^2}{\sum x_i^2} \right)$$

$$= \frac{\sigma^2}{\sum x_i^2} (a^2 + 2ab) + \frac{\sigma^2}{(\sum x_i)^2} \cdot b^2 n$$

$$= \sigma^2 \left[\frac{a^2 + 2ab}{\sum x_i^2} + \frac{b^2 n}{(\sum x_i)^2} \right]$$

(6)

(i)

Given model is multiple
linear regression model as we
have 2 independent variables
(TV & Radio).

(ii)

$$\hat{y} = 7.79108 + 0.22207(TV) + 0.07042(Radio) \quad (1)$$

(iii)

$$\hat{y}(TV=85 \text{ \& Radio}=30)$$

$$= 7.79108 + 0.22207 \times 85 + 0.07042 \times 30 \quad (2)$$

$$= 28.77963 \text{ (in thousands of units)}$$

(iv)

$$\text{at } \alpha = 0.05$$

we see

$$P_{TV} = 0.0145$$

(P-value for TV)

Since $P_{TV} < \alpha \Rightarrow \text{Reject } H_0$ (Reject the null hypothesis)

that TV is not a significant predictor

$\Rightarrow$ TV is a significant predictor

$$P_{Radio} = 0.6107$$

$$\text{Since } \alpha \geq P_{Radio}$$



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we fail to Reject H_0
(Fail to Reject Null hypothesis
that Radio is significant predictor
for Sales)

$\Rightarrow$ Radio is not a significant
Predictor (1)

Extra work for Q5(ii)

We know :

$$(\sum x_i)^2 \leq \sum (x_i^2)$$

$$\left(\frac{\sum x_i}{n}\right)^2 \leq \frac{1}{n^2} \sum (x_i^2)$$

$$n^2(\bar{x})^2 < \sum (x_i^2)$$

$$\frac{1}{n^2(\bar{x})^2} > \frac{1}{\sum (x_i^2)} \Rightarrow \frac{\sigma^2}{n^2(\bar{x})^2} > \frac{\sigma^2}{\sum (x_i^2)}$$

$$= \frac{1}{n} \cdot \frac{\sigma^2}{n(\bar{x})^2} > \frac{\sigma^2}{\sum (x_i^2)} \Rightarrow \frac{1}{n} \text{Var}(\hat{\beta}_2) > \text{Var}(\hat{\beta}_1)$$

$\Rightarrow \text{Var}(\hat{\beta}_1) < \text{Var}(\hat{\beta}_2)$

